

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q23-2290

Hastings 138 kV
December 29, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2290. The Interconnection Customer proposes a 100.2 MW solar photovoltaic facility interconnecting at the Hastings 138 kV substation in the Pennsylvania area. The request was received on February 18, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 752 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$94,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

MCCLELLAN-ADC - CHANNEL ENERGY CTR 138kV: 102.9% loading (314 MVA) following the outage of the CO LINE RD SUBSTATION - TALENENERGY MARTINS CRK LLC HARRISBURG 500kV; DFAX = 5.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MCCLELLAN-ADC - CHANNEL ENERGY CENTER 138kV	Reconductor line	\$232,000	41% (\$94,000)
	Total allocated to project		\$94,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.